Modeling Arbitrage Opportunities in Tri-Listed Stocks Through Event-Based Process Mining

Leonardo Pratelli*

University of Pisa and Scuola Superiore Sant'Anna

Abstract

Financial markets increasingly exhibit complex microstructural dynamics, especially for assets listed across multiple exchanges. Tri-listed stocks, traded simultaneously in different geographic and currency environments, provide a natural setting for the emergence of short-lived arbitrage opportunities due to price discrepancies, latency effects, and exchange-specific liquidity conditions. This paper proposes a novel framework that integrates high-frequency financial data with process mining techniques to analyze the lifecycle of arbitrage opportunities. By transforming market events into structured event logs, we model arbitrage episodes as processes composed of distinct phases, such as detection, spread expansion, execution, and convergence. Using tri-listed equities (e.g., Unilever) as a case study, we apply process discovery algorithms (e.g., Alpha algorithm and Petri nets) to uncover recurrent behavioral patterns in arbitrage formation and resolution. The approach enables the identification of dominant process variants, timing structures, and inefficiencies in market alignment across exchanges. The results demonstrate how process mining can provide a new lens for understanding financial market microstructure, offering insights into the temporal dynamics and structural patterns underlying arbitrage opportunities. This work contributes to bridging data science, financial economics, and process mining, opening new directions for the analysis of high-frequency trading phenomena.Vedi la modifica? Ora? Ora? Sì vedi la modifica, grazie!

1. Introduction

Introduction part...

2. Background

Math part....

*l.pratelli3@studenti.unipi.it; s3l.pratelli@santannapisa.it

2.1 The Mechanics of Modern Electronic Markets

Law part...

3. Conclusion

Something else...